

Ryan Pham

Fountain Valley, CA | ryandphambusiness@gmail.com | [GitHub](#) | [Portfolio](#)

EDUCATION

California State Polytechnic University, Pomona

Expected Graduation Date: December 2026

B.S. Computer Science

GPA: 3.92

- Organizations: Google Developer Student Clubs, Computer Science Society, Data Science & AI Club, Bronco Space Lab
- Honors: President's List (2x), Dean's List (4x)

SKILLS

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL

Technologies: React.js, FastAPI, Node.js, Git, PostgreSQL, MongoDB, SQLite, PyTorch, F Prime (F')

EXPERIENCE

CADENCE-SWANS CubeSat (AFRL University Nanosatellite Program)

Pomona, CA

Flight Software Engineer

May 2026 - Present

- Engineered the Real-Time Clock (RTC) Manager in C++ using NASA's F Prime (F') framework for a 12U CubeSat, defining ports, commands, telemetry channels, and interfaces for spacecraft timekeeping.
- Architecting the flight software topology by mapping F Prime components to planned spacecraft hardware and establishing interfaces between avionics, software components, and mission subsystems under UNP requirements.
- Configured the RTC Manager within the F Prime system topology, linking supporting components and preparing the design for future hardware integration, verification, and system-level testing.

CIC | PCUBED Data Science Summer Research Project

Fullerton, CA

Research Scholar

June 2023 - August 2023

- Trained machine learning models with Python and scikit-learn on 28K+ police records, engineering temporal, spatial, and demographic features to achieve 80% forecasting accuracy on New York City shooting incident data.
- Processed and standardized inconsistent real-world datasets through data cleaning, transformation, and feature preparation to improve data quality and experiment reproducibility.

Code Ninjas

Fountain Valley, CA

Programming Instructor

March 2023 - August 2024

- Instructed 100+ students in Python, JavaScript, Java, and C# through hands-on, project-based programming curricula spanning elementary through high school levels.
- Directed development camps using Unity/C#, Roblox Studio/Lua, and TinkerCAD, maintaining a 95% student retention rate through interactive technical instruction.

NASA L'SPACE Mission Concept Academy

Remote

Team Member

June 2023 - August 2023

- Collaborated within a 13-person interdisciplinary team to formulate a Mars mission concept for investigating crustal magnetic fields in the Southern Highlands region.
- Evaluated cost and technical feasibility for a \$300M Discovery-class mission concept, contributing to computer hardware, payload, and dual-rover deployment planning.

PROJECTS

UAV Drone Tracker | Python, YOLO, ByteTrack, OpenCV, NumPy

- Engineered a real-time multi-UAV computer vision pipeline using YOLO, ByteTrack, and OpenCV, associating detections across video frames with persistent track IDs and independent state histories for each UAV.
- Applied Kalman filtering and motion estimation to position and velocity states, smoothing noisy detections and maintaining continuously updated motion estimates for individual UAV tracks.
- Designed multi-horizon trajectory forecasting at 0.5, 1.0, and 2.0 seconds, projecting future UAV positions from filtered motion states and maintaining prediction histories for subsequent evaluation.
- Recorded per-UAV telemetry and motion histories by track ID, preserving time-series state data for real-time visualization, trajectory analysis, and prediction evaluation.

CineMatch | Python, FastAPI, React, OpenAI API, MCP, TMDB API

- Developed a full-stack agentic AI application spanning 3 services — React frontend, FastAPI backend, and MCP tool server — using REST APIs, SSE, and OpenAI tool calling for personalized movie and TV recommendations.
- Built a Model Context Protocol (MCP) server with 3 schema-validated tools routing 100% of TMDB access through a single layer; required per-title detail/trailer tool calls so recommendations use verified metadata (posters, runtime, ratings) instead of LLM hallucination.
- Reduced TMDB search/discover latency by ~50% via parallel async HTTP (httpx, asyncio.gather) across movie and TV endpoints; implemented an agent loop supporting up to 24 concurrent tool-call rounds with structured JSON output.

Buzzr | JavaScript, Node.js, SQLite, Twilio API, Discord.js

- Deployed a Node.js Discord application serving 100+ users across 2 sports communities, automating announcements and event-driven notification workflows.
- Integrated the Twilio API to process and deliver 10+ daily SMS alerts, extending server notifications to subscribed mobile users.